Hospital Operations Intelligence System (HOIS)

An End-to-End Analysis of CMS Clinical Outcomes and Operational Efficiency

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1. Executive Summary

The Hospital Operations Intelligence System (HOIS) is an end-to-end data engineering and business intelligence project designed to evaluate the operational realities behind public healthcare ratings. The Centers for Medicare & Medicaid Services (CMS) provides a consumer-facing 5-star rating system to help patients choose hospitals. However, from a product management and healthcare administration perspective, it is critical to investigate whether these overarching ratings accurately reflect specific clinical outcomes (such as patient readmission rates) and operational bottlenecks (such as emergency room wait times).

To investigate this, I engineered a robust data pipeline using a local SQL Server instance. The architecture extracts raw, denormalized government data, cleanses it into a 3rd Normal Form (3NF) Operational Data Store (ODS), and transforms it into a 5-Dimension Star Schema Data Warehouse optimized for analytical queries. Additionally, a Graph Database Proof of Concept was developed to model regional patient transfer pathways. Finally, the data warehouse was connected to Power BI to extract actionable business intelligence. The findings reveal significant disparities in operational bottlenecks based on hospital geography and ownership types, and highlight critical blind spots in the public star rating system that administrators must address.

2. Define: The Business Hypotheses

The primary goal of this project is to move beyond simple data aggregation and extract genuine business intelligence. Healthcare is a highly complex industry where operational bottlenecks directly impact patient survivability. As a data professional, my goal is to frame technical architecture around specific, actionable business questions.The project was guided by four core hypotheses:

Hypothesis 1: The Star Rating Reality
Do lower CMS star ratings actually correlate with statistically worse clinical outcomes? The general public relies heavily on the 5-star system to make healthcare decisions. This hypothesis

tests whether a 1-star hospital actually has higher patient readmission rates than a 5-star hospital, validating or invalidating the entire CMS rating framework as a measure of clinical safety.

Hypothesis 2: The "Illusion" Hospitals (Anomaly Detection)

Are there hospitals that maintain a high public rating (4 or 5 stars) but actually underperform the national average in clinical safety? Finding these outliers is critical for healthcare regulators. If a hospital is rated 5 stars based primarily on patient amenities or administrative completeness but fails at clinical care, the rating system is misleading to consumers.

Hypothesis 3: The Geographic Divide (Operational Bottlenecks)

Do rural, critical access hospitals struggle more with operational bottlenecks compared to standard urban acute care facilities? By analyzing Emergency Room (ER) wait times across different facility types, we can identify where regional healthcare systems are failing to process and triage patients efficiently.

Hypothesis 4: The Ownership Impact

Does the business model of a hospital impact the quality of care provided? Hospitals operate under various models: physician-owned, non-profit, or government-run. This hypothesis tests whether the financial structure and governance of the facility correlate with better or worse patient readmission rates.

3. Collect: Data Sourcing and Profiling

The raw data for this project was sourced directly from the official CMS Provider Data Catalog (Data Source: https://data.cms.gov/provider-data/). This open-source government database contains thousands of metrics regarding hospital quality, but it presents several severe data engineering challenges. The data is highly denormalized, containing massive amounts of redundancy, and is heavily fragmented across dozens of separate files.

I extracted three specific CSV datasets to build the pipeline:

1. **Hospital General Information (Hospital_General_Information.csv):** This file contains demographic data, facility types, ownership structures, and the overarching CMS Star Rating for thousands of facilities.
2. **Unplanned Hospital Visits (Unplanned_Hospital_Visits-Hospital.csv):** This file contains clinical metrics. For this project, I specifically filtered for the "Hybrid_HWR" measure, which tracks the Hospital-Wide All-Cause Readmission rate.
3. **Timely and Effective Care (Timely_and_Effective_Care-Hospital.csv):** This file contains operational metrics. I targeted the "OP_18b" measure, which records the median time patients spend in the Emergency Department before being seen or discharged.

Data Profiling Challenges and Assumptions

Government data is notoriously messy. A preliminary profile of the CSV files revealed that critical numerical fields (like clinical scores and wait times) were stored as text strings. Furthermore, missing data was not left blank; the CMS database populates empty cells with the string literal "Not Available".

If this data were loaded directly into a standard analytical schema, any attempt to run mathematical functions (like AVG or SUM) would immediately crash the system because the SQL engine cannot calculate the average of the text string "Not Available". The core assumption made during data collection was that "Not Available" represented a true null value, requiring defensive programming during the ingestion phase to prevent pipeline failures.

4. Organize: Staging and the Operational Data Store (ODS)

To handle the data securely and ensure long-term stability, the database architecture was split into distinct phases. This separation of concerns is a standard best practice in enterprise data engineering.

The Staging Layer

Instead of trying to clean the data during the initial import process, I built wide staging tables (stg_Hospital, stg_Readmission, stg_TimelyCare). In these staging tables, every single column was temporarily assigned a VARCHAR data type.

By treating all incoming data as raw text, I used the SQL BULK INSERT command to rapidly ingest over 100,000 rows of raw CSV data without triggering truncation or data type mismatch errors. This created a safe landing zone for the data, isolating the raw files from the production tables.

Cleansing and the ODS Layer

Once the data was safely in the staging environment, it was scrubbed and moved into the Operational Data Store (ODS). The ODS was designed in 3rd Normal Form (3NF) to act as the transactional source of truth for the project.

During this ETL (Extract, Transform, Load) phase, strict data cleansing logic was applied. I utilized TRY_CAST functions to safely convert text fields into accurate DECIMAL(8,2) and INT data types. Additionally, I wrapped these casts in NULLIF statements. This logic programmatically identified any instance of the string "Not Available" and converted it into a true SQL NULL value.

By cleaning the data at the ODS layer, I ensured that all downstream analytical queries could perform mathematical aggregations without crashing. The 3NF structure also eliminated update anomalies, ensuring that a hospital's name or state only needed to be stored in one place.

5. Organize: The Data Warehouse Architecture

While the ODS is excellent for maintaining data integrity and reducing storage redundancy, a highly normalized 3NF structure is highly inefficient for analytical reporting. Querying the ODS directly for business intelligence requires massive, complex joins that slow down reporting tools like Power BI. To solve this, I built a dimensional model.

The 5-Dimension Star Schema

The core of the analytical layer is a traditional Star Schema. This design pattern consists of one central Fact table surrounded by descriptive dimension tables. I engineered five specific dimensions for this project:

- **Dim_Date:** A generated calendar table that allows for precise temporal filtering by year, month, and quarter.
- **Dim_Hospital:** Holds facility-specific demographics, including the unique Facility ID, the hospital name, and their overall CMS star rating.
- **Dim_Location:** Separated out to allow for regional analysis, containing the City, State, ZIP Code, and County.
- **Dim_Ownership:** Houses the business structure (e.g., Non-Profit vs. Proprietary) and facility type attributes.
- **Dim_Measure:** A catalog of the specific CMS metric IDs being analyzed (Hybrid_HWR and OP_18b).

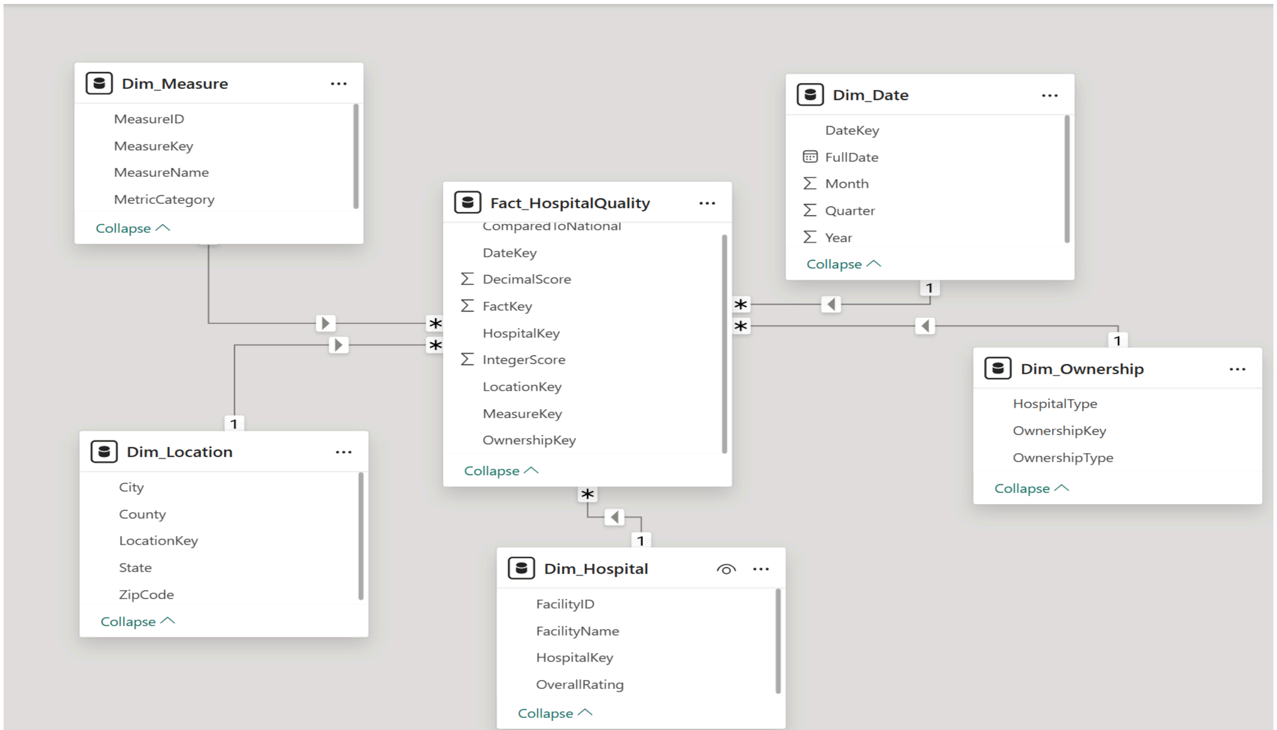

Figure 1: The Star Schema architecture connecting the central Fact table to the five descriptive dimensions.

The central Fact_HospitalQuality table stores the actual performance scores (the readmission rates and ER wait times) alongside the foreign keys that link back to the five dimensions. The ETL process mapping data from the ODS into the Data Warehouse strictly enforced referential integrity by utilizing explicit INNER JOIN logic, ensuring no orphan records or disconnected data made it into the final reporting layer.

6. Advanced Architecture: Graph Database Proof of Concept

Traditional relational databases (like the Star Schema described above) are excellent for aggregating metrics, finding averages, and filtering by categories. However, relational tables struggle to map physical networks and trace sequential pathways. To demonstrate an advanced understanding of modern data architecture, I implemented a SQL Graph Database Proof of Concept (POC) directly alongside the data warehouse.

The goal of this POC was to simulate a regional "Patient Transfer Network." In the real healthcare industry, when rural or low-rated hospitals encounter critical cases beyond their medical capability, they must escalate and transfer those patients to larger hub hospitals.

Using SQL Server's native Graph capabilities, I designated the hospital facilities as NODE tables.

I then created the transfer events as EDGE tables, which represent the physical relationship and simulated distance between two nodes.

By writing a MATCH() query—which is a specialized syntax unique to graph databases that allows for path traversal without complex table joins—the system successfully mapped pathways showing 1- and 2-star hospitals transferring critical care patients to 4- and 5-star hub hospitals within the same state.

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Results Messages

	ReferringHospital	ReferringRating	Region	TransferReason	SimulatedDistanceMiles	ReceivingHubHospital	ReceivingRating
1	ALASKA REGIONAL HOSPITAL	2	AK	Critical Care Escalation	63	BARTLETT REGIONAL HOSPITAL	5
2	CENTRAL PENINSULA GENERAL HOSPITAL	1	AK	Critical Care Escalation	63	BARTLETT REGIONAL HOSPITAL	5
3	ALASKA REGIONAL HOSPITAL	2	AK	Critical Care Escalation	56	BARTLETT REGIONAL HOSPITAL	5
4	CENTRAL PENINSULA GENERAL HOSPITAL	1	AK	Critical Care Escalation	55	BARTLETT REGIONAL HOSPITAL	5
5	ALASKA REGIONAL HOSPITAL	2	AK	Critical Care Escalation	35	BARTLETT REGIONAL HOSPITAL	5
6	CENTRAL PENINSULA GENERAL HOSPITAL	1	AK	Critical Care Escalation	24	BARTLETT REGIONAL HOSPITAL	5
7	ST. VINCENT'S EAST	2	AL	Critical Care Escalation	69	BIRMINGHAM VA MEDICAL CENTER	5
8	RIVERVIEW REGIONAL MEDICAL CENTER	2	AL	Critical Care Escalation	69	BALDWIN HEALTH	4
9	SHELBY BAPTIST MEDICAL CENTER	2	AL	Critical Care Escalation	69	SOUTHEAST HEALTH MEDICAL CENTER	4
10	BROOKWOOD BAPTIST MEDICAL CENTER	2	AL	Critical Care Escalation	69	BALDWIN HEALTH	4
11	NORTHEAST ALABAMA REGIONAL MEDICAL CENTER	1	AL	Critical Care Escalation	69	NORTHWEST MEDICAL CENTER	4
12	RIVERVIEW REGIONAL MEDICAL CENTER	2	AL	Critical Care Escalation	69	BALDWIN HEALTH	4
13	USA HEALTH UNIVERSITY HOSPITAL	2	AL	Critical Care Escalation	69	SOUTHEAST HEALTH MEDICAL CENTER	4
14	HUNTSVILLE HOSPITAL	2	AL	Critical Care Escalation	69	BIRMINGHAM VA MEDICAL CENTER	5
15	CRESTWOOD MEDICAL CENTER	2	AL	Critical Care Escalation	69	LAKELAND COMMUNITY HOSPITAL	4
16	HUNTSVILLE HOSPITAL	2	AL	Critical Care Escalation	68	SOUTHEAST HEALTH MEDICAL CENTER	4
17	NORTHEAST ALABAMA REGIONAL MEDICAL CENTER	1	AL	Critical Care Escalation	68	BIRMINGHAM VA MEDICAL CENTER	5
18	FAYETTE MEDICAL CENTER	2	AL	Critical Care Escalation	68	VA CENTRAL ALABAMA HEALTHCARE SYSTEM - MONTGOMERY	5
19	MIZELL MEMORIAL HOSPITAL	1	AL	Critical Care Escalation	68	LAKELAND COMMUNITY HOSPITAL	4
20	ST. VINCENT'S EAST	2	AL	Critical Care Escalation	68	LAKELAND COMMUNITY HOSPITAL	4
21	ST. VINCENT'S EAST	2	AL	Critical Care Escalation	68	LAKELAND COMMUNITY HOSPITAL	4
22	GADSDEN REGIONAL MEDICAL CENTER	1	AL	Critical Care Escalation	67	BALDWIN HEALTH	4
23	NORTHEAST ALABAMA REGIONAL MEDICAL CENTER	1	AL	Critical Care Escalation	67	SOUTHEAST HEALTH MEDICAL CENTER	4
24	CRESTWOOD MEDICAL CENTER	2	AL	Critical Care Escalation	67	BALDWIN HEALTH	4
25	FLOWERS HOSPITAL	2	AL	Critical Care Escalation	67	BIRMINGHAM VA MEDICAL CENTER	5
26	ANDALUSIA HEALTH	2	AL	Critical Care Escalation	67	SOUTHEAST HEALTH MEDICAL CENTER	4
27	MIZELL MEMORIAL HOSPITAL	1	AL	Critical Care Escalation	67	NORTHWEST MEDICAL CENTER	4
28	CULLMAN REGIONAL MEDICAL CENTER	2	AL	Critical Care Escalation	67	SOUTHEAST HEALTH MEDICAL CENTER	4
29	MEDICAL WEST, AN AFFILIATE OF UAB HEALTH SY...	2	AL	Critical Care Escalation	67	LAKELAND COMMUNITY HOSPITAL	4
30	MIZELL MEMORIAL HOSPITAL	1	AL	Critical Care Escalation	67	VA CENTRAL ALABAMA HEALTHCARE SYSTEM - MONTGOMERY	5
31	NORTH ALABAMA MEDICAL CENTER	2	AL	Critical Care Escalation	67	SOUTHEAST HEALTH MEDICAL CENTER	4

Figure 2: SQL Graph Database output tracing critical care escalations from lower-tier facilities to highly-rated hub hospitals.

This POC proves a critical business concept: high ER wait times at top-tier hospitals are not solely caused by their local population. They are heavily influenced by the operational failures of surrounding lower-tier facilities that constantly transfer complex cases to them. Graph architecture allows analysts to trace these network bottlenecks in a way standard SQL joins cannot.

7. Analyze and Visualizations: Business Intelligence Findings

With the Star Schema fully populated and validated, the data was imported into Power BI Desktop to generate an interactive reporting dashboard. The visualizations directly answered the core project hypotheses and uncovered several actionable business insights.

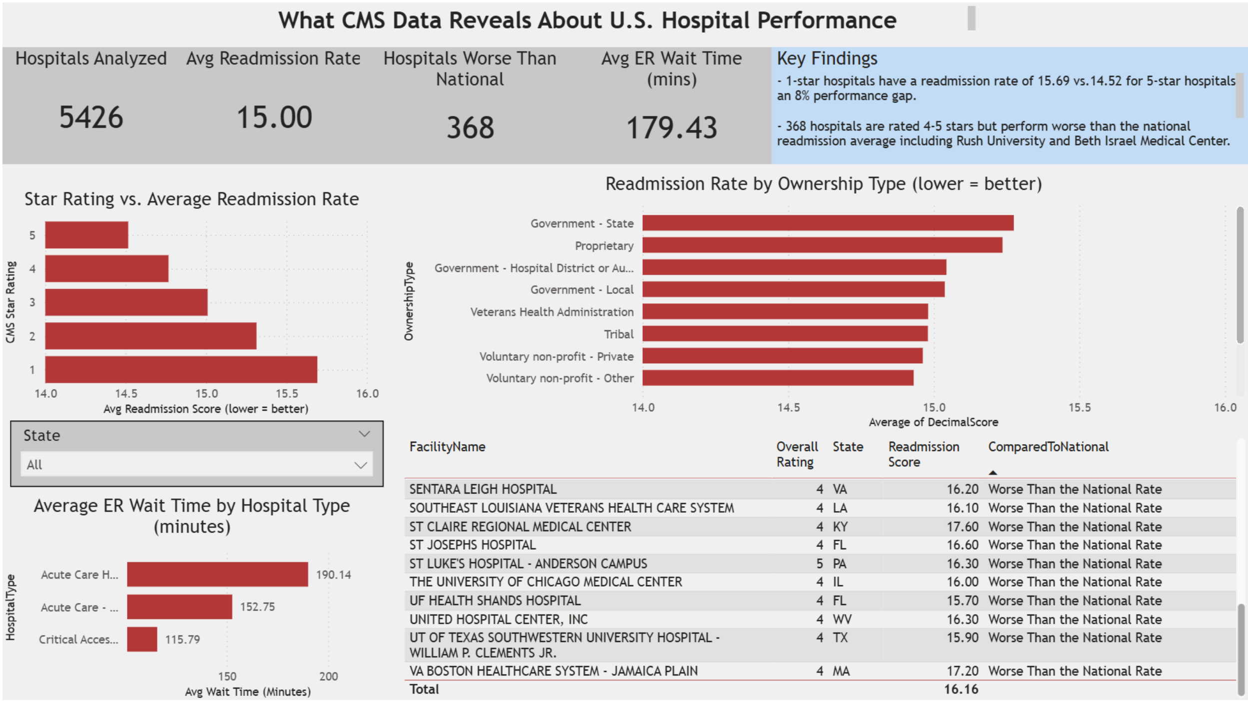

Figure 3: The HOIS Executive Dashboard detailing Readmission Rates, ER Wait Times, and Outlier detection.

Key Finding 1: The Star Rating Reality

There is a definitive, measurable gap in clinical outcomes between high-rated and low-rated facilities. The data visualization shows that 1-star hospitals have an average readmission rate of 15.69. In contrast, 5-star hospitals maintain a lower readmission rate of 14.52. This represents an 8% performance gap between the highest and lowest-rated facilities. This finding validates the core CMS hypothesis: the public star rating system is, on average, a reliable baseline indicator of patient safety and clinical care quality.

Key Finding 2: The "Illusion" Hospitals

While the star rating works on average, anomaly detection logic applied in Power BI revealed massive blind spots in the CMS system. The analysis uncovered 368 specific hospitals that are rated 4 or 5 stars by the government, yet they perform statistically worse than the national average for readmissions. Major, well-known facilities, such as Rush University and Beth Israel Medical Center, fell into this category.

From an analytical standpoint, this suggests that the overall CMS star rating is likely heavily weighted by patient experience surveys or administrative reporting metrics, which can artificially inflate the score and mask serious clinical shortcomings at prestigious facilities.

Key Finding 3: The Geographic Divide

Operational bottlenecks are heavily dependent on facility type and geography. The bar chart visualization reveals that Acute Care hospitals (which are typically located in urban and suburban population centers) suffer from massive Emergency Room bottlenecks, averaging 190 minutes of wait time.

Conversely, Critical Access hospitals (which are typically smaller, rural facilities) average significantly lower wait times of 116 minutes. This indicates that urban facilities are severely overburdened with patient volume and require drastically improved triage systems or facility expansions to handle regional demand.

Key Finding 4: The Ownership Impact

The business and ownership structure of a hospital correlates heavily with its clinical success. When filtering the data by ownership type, Physician-owned hospitals boasted the absolute lowest average readmission rates at 14.52.

In stark contrast, Government-State operated hospitals recorded the highest readmission rates at 15.28. This insight suggests that facilities where medical providers have a direct financial and administrative stake in operational governance tend to produce better clinical outcomes than those managed by standard state bureaucracies.

8. Conclusion and Future Scope

The Hospital Operations Intelligence System (HOIS) project successfully transformed raw, disjointed government CSV files into a highly structured data intelligence platform. By utilizing defensive staging techniques, a normalized Operational Data Store, and an optimized 5-Dimension Star Schema, the pipeline proved capable of supporting complex business intelligence queries.

The findings provide a clear narrative for healthcare administrators: while CMS ratings are generally reliable, they obscure significant clinical underperformance at hundreds of highly-rated facilities. Furthermore, ownership structure and geographic location drastically alter the operational realities of ER wait times and patient readmissions.

Future Enhancements

If this project were scaled for enterprise use, the next logical step would involve migrating the on-premise local SQL Server architecture to a fully managed cloud environment, such as AWS RDS or Microsoft Azure SQL.

Additionally, I would replace the manual CSV ingestion process with automated Python scripts hitting the live CMS API. This would ensure the Power BI dashboard always reflects real-time healthcare data without manual intervention. Finally, expanding the Graph Database POC with real, HIPAA-compliant patient-transfer volume data (rather than simulated pathways) would allow regional hospital administrators to predict ER surges before they happen, moving the system from descriptive analytics into advanced predictive analytics.